

DANDONG, China

WMO: 544970

Lat: 40.0317N

Lon: 124.3275E

Elev: 46

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.0	(c) 6.5	(d) -18.5	(e) 2.0	(f) 9.6	(g) -13.9	(h) 2.6	(i) 12.6	(j) 21.0	(k) 19.2	(l) 19.3	(m) 19.0	(n) 6.6	(o) 340	(p) 0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.5	(c) 86.4	(d) 75.2	(e) 84.0	(f) 73.9	(g) 81.9	(h) 72.9	(i) 79.0	(j) 82.8	(k) 77.4	(l) 80.8	(m) 76.0	(n) 79.1	(o) 6.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.9	145.3	81.0	76.4	137.8	79.4	75.0	131.3	78.0	42.7	82.9	41.1	81.2	39.6	79.1	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	16.4	14.3	-1.4	90.6	4.9	2.6	-4.9	92.5	-7.8	94.1	-10.6	95.6	-14.1	97.5
			-2.8	80.7	4.6	2.3	-6.1	82.4	-8.8	83.7	-11.4	85.0	-14.8	86.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	20.0	26.3	36.7	49.4	60.1	68.8	74.8	75.4	67.0	53.7	37.8	23.7
	DBStd	20.28	7.19	7.27	6.20	5.90	4.57	3.89	3.32	3.58	4.81	6.76	8.80	7.62
	HDD50	3319	931	663	413	81	1	0	0	0	0	41	374	815
	HDD65	6478	1396	1083	876	469	162	11	0	0	34	352	815	1280
	CDD50	3168	0	0	1	61	314	563	768	788	509	156	8	0
	CDD65	855	0	0	0	0	10	124	303	323	93	2	0	0
	CDH74	4636	0	0	0	3	85	536	1571	2056	376	9	0	0
	CDH80	920	0	0	0	0	9	92	338	452	29	0	0	0
Wind	WSAvg	6.1	6.9	6.8	6.9	6.8	5.9	5.0	4.8	5.0	5.4	6.0	6.9	6.9
Precipitation	PrecAvg	38.1	0.4	0.4	0.7	2.1	2.6	4.4	11.3	9.1	3.7	1.7	1.1	0.5
	PrecMax	58.0	2.4	1.6	2.1	5.7	5.7	10.5	25.7	20.0	7.7	6.7	2.4	1.3
	PrecMin	26.4	0.0	0.0	0.1	0.3	0.4	0.8	3.9	1.5	0.7	0.0	0.2	0.0
	PrecStd	8.3	0.4	0.4	0.5	1.4	1.5	2.5	5.7	5.0	1.9	1.5	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	48.1	58.6	73.0	81.0	86.3	90.3	90.1	83.3	74.4	61.7	45.1
		MCWB	34.5	38.7	45.0	56.3	62.6	70.0	79.0	78.9	71.9	63.0	55.8	41.7
	2%	DB	36.3	43.0	53.0	67.2	76.0	82.6	86.6	86.6	80.2	69.9	57.6	40.7
		MCWB	31.6	35.3	42.2	52.7	60.2	67.5	75.9	76.0	68.9	60.0	52.1	36.5
	5%	DB	33.4	39.7	49.6	63.2	72.9	79.6	83.6	84.6	78.1	66.9	54.4	37.5
		MCWB	28.9	34.5	40.4	50.6	59.1	67.3	74.7	75.0	67.4	58.1	49.2	33.5
	10%	DB	30.7	37.1	46.6	59.9	69.9	76.9	81.1	82.4	76.0	64.2	50.9	34.8
		MCWB	26.3	32.1	39.1	49.2	58.1	67.2	74.0	74.4	66.4	56.6	45.8	30.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	40.3	48.6	59.8	66.3	73.4	81.6	81.9	75.5	66.1	57.2	42.8
		MCDB	38.6	43.9	54.8	67.0	75.0	82.0	87.2	85.6	80.0	70.7	59.5	44.8
	2%	WB	32.5	37.5	44.9	55.6	63.7	71.3	78.9	79.4	72.9	63.0	53.8	37.1
		MCDB	35.4	41.0	49.3	62.0	71.1	77.0	83.2	82.8	76.9	67.2	56.5	39.5
	5%	WB	29.5	35.2	42.6	53.5	61.9	70.1	77.1	78.1	71.0	60.8	50.2	34.2
		MCDB	33.0	39.2	47.1	60.5	68.4	75.5	80.5	81.5	75.0	64.7	53.4	36.8
	10%	WB	26.6	32.7	40.5	51.2	60.4	69.0	75.6	76.9	69.2	58.5	46.8	31.3
		MCDB	30.2	36.5	45.5	57.3	66.8	74.1	78.6	80.1	73.3	62.4	50.1	34.2
Mean Daily Temperature Range	5% DB	MDBR	14.8	15.5	15.6	16.6	16.2	12.9	10.1	12.5	16.6	17.7	14.6	13.5
		MCDBR	16.3	17.4	19.4	21.7	21.6	19.0	15.4	15.9	18.2	19.5	17.3	15.1
		MCWBR	12.6	12.0	11.4	10.8	9.6	7.7	6.5	7.0	8.5	10.9	12.7	12.2
	5% WB	MCDBR	15.5	15.4	16.9	18.0	17.2	14.1	11.1	11.1	14.2	16.0	15.3	14.1
		MCWBR	13.0	11.9	11.5	10.4	9.0	6.8	5.8	6.3	8.3	10.8	13.0	12.8
Clear-Sky Solar Irradiance	taub	0.402	0.478	0.538	0.624	0.652	0.734	0.710	0.604	0.530	0.556	0.489	0.415	
	taud	1.962	1.813	1.735	1.614	1.611	1.526	1.614	1.820	1.918	1.762	1.853	1.951	
	Ebn at Noon	224	222	224	215	214	197	200	218	225	195	191	206	
	Edh at Noon	41	55	67	80	82	89	81	64	55	58	46	39	
All-Sky Solar Radiation	RadAvg	765	1006	1373	1585	1773	1567	1329	1431	1405	1052	733	651	
	RadStd	50	72	99	126	146	192	140	166	114	67	76	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page